CSCI 49376 Assignment 1

Mia Elena Yábar (Partner A: design) · Jas (Partner B: implementation)
Big Data Technology · Prof. Alex Liew · This document: deliverables 1–10

1. Name of the selected dataset

Diabetes 130-US Hospitals for Years 1999–2008 (Kaggle: *brandao/diabetes*). CSV file used: `diabetic_data.csv`.

2. Description of the original dataset

Ten years of inpatient stays by diabetic patients at 130 U.S. hospitals (Strack et al., 2014). Each row is one hospital stay: the patient, how they were admitted and discharged, what was done during the stay, up to three diagnoses, two lab results, and whether they were readmitted.

- **Possible keys:** `encounter_id` is unique on every row. `patient_nbr` repeats (71,518 patients across 101,766 stays), so it identifies a separate patient. The three `*_id` columns are codes for small reference lists, and `diag_1`–`diag_3` hold 915 distinct diagnosis codes.
- **Missing values:** written as `?` or `None`; all are stored as NULL. `weight` 96.9% · `max_glu_serum` 94.7% · `A1Cresult` 83.3% · `medical_specialty` 49.1% · `payer_code` 39.6% · `race` 2.2% · `diag_3` 1.4% · `diag_2` 0.4% · `gender` 3 rows.
- **Inconsistencies:** 123 patients have two different race values and 3 have two genders across their stays (we keep the value from the patient's latest stay). Age changes between stays for 1,574 patients, so age belongs to the stay, not the patient.

3. List and description of the original attributes

Attribute	Description
`encounter_id`	Unique ID of one hospital stay
`patient_nbr`	Unique ID of a patient (a patient can have several stays)
`race`	Caucasian, AfricanAmerican, Hispanic, Asian, Other
`gender`	Female, Male, Unknown/Invalid
`age`	Age group in 10-year bands, e.g. [70-80)
`weight`	Weight band in pounds
`admission_type_id`	Code: type of admission (e.g. emergency, elective)
`discharge_disposition_id`	Code: where the patient went at discharge (e.g. home)
`admission_source_id`	Code: where the patient came from (e.g. referral, ER)
`time_in_hospital`	Days in the hospital (1–14)
`payer_code`	Insurance code (e.g. MC = Medicare)
`medical_specialty`	Specialty of the admitting doctor
`num_lab_procedures`	Lab tests during the stay
`num_procedures`	Other procedures during the stay

num_medications	Distinct medications given
number_outpatient	Outpatient visits in the prior year
number_emergency	Emergency visits in the prior year
number_inpatient	Inpatient visits in the prior year
diag_1	Primary diagnosis (ICD-9 code)
diag_2	Secondary diagnosis (ICD-9 code)
diag_3	Additional diagnosis (ICD-9 code)
number_diagnoses	Number of diagnoses recorded
max_glu_serum	Glucose test result: >200, >300, Norm, None (not measured)
A1Cresult	HbA1c test result: >7, >8, Norm, None (not measured)
diabetesMed	Any diabetes medication prescribed: Yes/No
readmitted	NO, >30 (after 30 days), <30 (within 30 days)
metformin, repaglinide, nateglinide, chlorpropamide, glimepiride, acetohexamide, glipizide, glyburide, tolbutamide, pioglitazone, rosiglitazone, acarbose, miglitol, troglitazone, tolazamide, examide, citoglipton, insulin, glyburide-metformin, glipizide-metformin, glimepiride-pioglitazone, metformin-rosiglitazone, metformin-pioglitazone, change	**Ignored, as instructed.** 23 medication columns (No, Steady, Up, Down) and change (whether any diabetes medication changed).

4. Original dataset dimensions

101,766 rows × 50 columns. 24 columns are ignored, so the design uses **26**.

5. Normalization process

Original table

```
DIABETIC_DATA(encounter_id, patient_nbr, race, gender, age, …, diag_1, diag_2, diag_3, …, readmitted)
```

Problem: diag_1, diag_2, diag_3 are a **repeating group**, three columns for the same kind of value.

First Normal Form (1NF)

Rule: one value per cell, no repeating groups. **Change:** each diagnosis gets its own row.

```
ENCOUNTER_1NF(encounter_id, diag_position, icd9_code, patient_nbr, race, gender, age, …, readmitted)
```

encounter_id	diag_position	icd9_code	time_in_hospital	…
1001	1	250.01	3	…

1001	2	401	3	…
1001	3	428	3	…

Key: (`encounter_id`, `diag_position`). The position is used instead of the code because 5,492 stays list the same code twice.
Problem: `time_in_hospital` and the other stay columns depend only on `encounter_id`, which is *part* of the key. This is a **partial dependency**, and it copies each stay's details once per diagnosis.

Second Normal Form (2NF)

Rule: 1NF, and every non-key column depends on the *whole* key. **Change:** move the stay columns into their own table.

```
ENCOUNTER(encounter_id, patient_nbr, age, time_in_hospital, …, readmitted)
ENCOUNTER_DIAGNOSIS(encounter_id, diag_position, icd9_code)
```

Why it is 2NF: ENCOUNTER has a one-column key, so nothing can depend on part of it. In ENCOUNTER_DIAGNOSIS, `icd9_code` needs both key columns.
Less redundancy: patient race and gender still repeat on every stay, so they move to PATIENT. The three code columns get reference tables, and DIAGNOSIS lists each code once. This removes 30,248 repeated copies of patient data, and every table stays in 2NF.

6. Relevant functional dependencies

#	Dependency	Used for
FD1	`encounter_id, diag_position` → `icd9_code`	ENCOUNTER_DIAGNOSIS
FD2	`encounter_id` → patient_nbr, age, weight, the three codes, time_in_hospital, … , readmitted	ENCOUNTER (the partial dependency removed in 2NF)
FD3	`patient_nbr` → race, gender	PATIENT
FD4	`admission_type_id` → description, and the same for discharge disposition and admission source	The three reference tables

7. Final list of entities and attributes

Entity (table)	Attributes	Rows
PATIENT	patient_nbr, race, gender	71,518
ENCOUNTER	encounter_id, patient_nbr, admission_type_id, discharge_disposition_id, admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty, num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted	101,766
ENCOUNTER_DIAGNOSIS	encounter_id, diag_position, icd9_code	303,496
DIAGNOSIS	icd9_code	915
ADMISSION_TYPE	admission_type_id, description	8
DISCHARGE_DISPOSITION	discharge_disposition_id, description	26

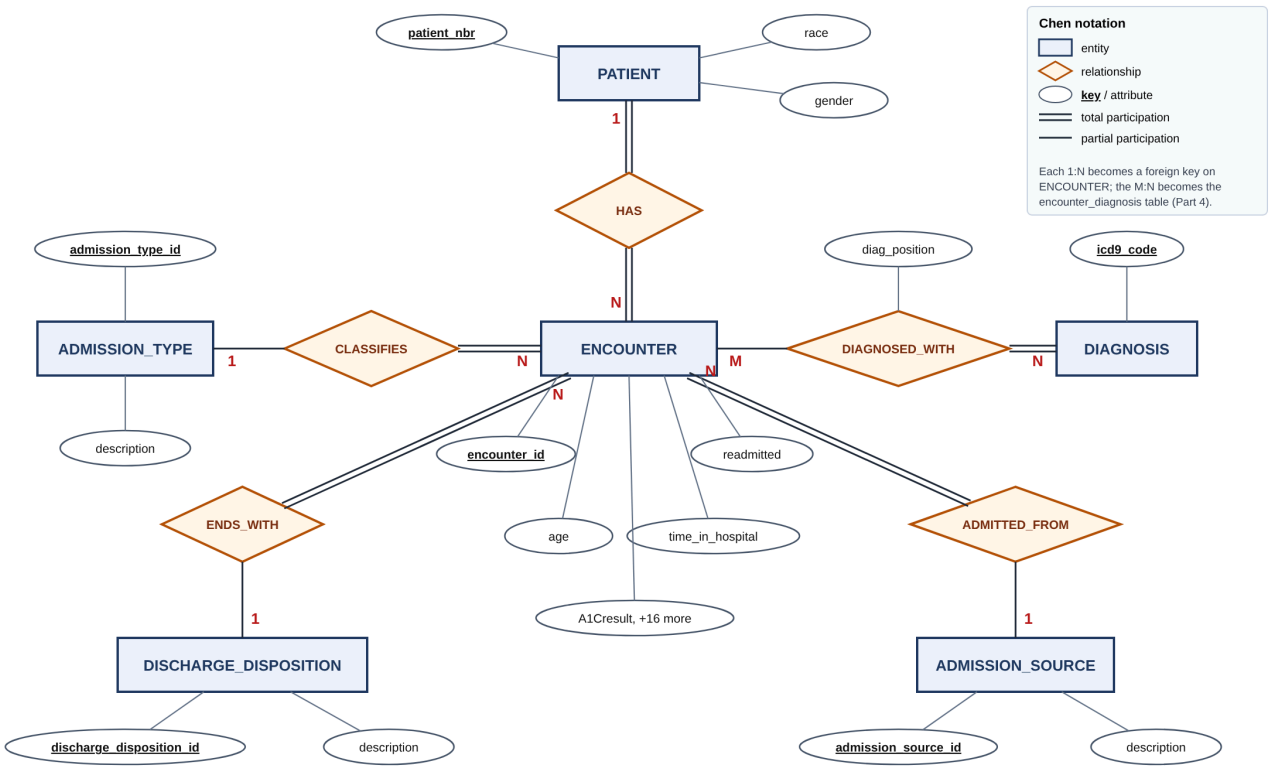
ADMISSION_SOURCE	admission_source_id, description	17

Descriptions for the three reference tables come from `IDs_mapping.csv`, published with the dataset (UCI). The row counts are the targets for checking the import.

8. Primary keys and foreign keys

Table	Primary key	Foreign keys
PATIENT	patient_nbr	none
ENCOUNTER	encounter_id	patient_nbr → PATIENT admission_type_id → ADMISSION_TYPE discharge_disposition_id → DISCHARGE_DISPOSITION admission_source_id → ADMISSION_SOURCE
ENCOUNTER_DIAGNOSIS	(encounter_id, diag_position)	encounter_id → ENCOUNTER icd9_code → DIAGNOSIS
DIAGNOSIS	icd9_code	none
ADMISSION_TYPE · DISCHARGE_DISPOSITION · ADMISSION_SOURCE	admission_type_id · discharge_disposition_id · admission_source_id	none

9. ER diagram with cardinalities and participations



Relationship	Cardinality	Participation
PATIENT *has* ENCOUNTER	1:N	Total on both sides
ADMISSION_TYPE *classifies* ENCOUNTER	1:N	Total for ENCOUNTER, partial for the type

DISCHARGE_DISPOSITION *ends with* ENCOUNTER	1:N	Total for ENCOUNTER, partial for the disposition
ADMISSION_SOURCE *admitted from* ENCOUNTER	1:N	Total for ENCOUNTER, partial for the source
ENCOUNTER *diagnosed with* DIAGNOSIS (attribute: diag_position)	M:N	Partial for ENCOUNTER (0–3 diagnoses), total for DIAGNOSIS

Chen notation: rectangle = entity, diamond = relationship, oval = attribute (key underlined), double line = total participation. There are no 1:1 relationships.

10. Final relational schema

Primary keys are **underlined**; foreign keys are in *italics*.

```
PATIENT(patient_nbr, race, gender)
ADMISSION_TYPE(admission_type_id, description)
DISCHARGE_DISPOSITION(discharge_disposition_id, description)
ADMISSION_SOURCE(admission_source_id, description)
DIAGNOSIS(icd9_code)
ENCOUNTER(encounter_id, patient_nbr, admission_type_id, discharge_disposition_id,
admission_source_id, age, weight, time_in_hospital, payer_code, medical_specialty,
num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency,
number_inpatient, number_diagnoses, max_glu_serum, A1Cresult, diabetesMed, readmitted)
ENCOUNTER_DIAGNOSIS(encounter_id, diag_position, icd9_code)
```

From the ER diagram: each 1:N relationship becomes a foreign key in ENCOUNTER, and the M:N relationship becomes the table ENCOUNTER_DIAGNOSIS. Create the tables in this order: PATIENT, the three reference tables and DIAGNOSIS, then ENCOUNTER, then ENCOUNTER_DIAGNOSIS.

Table diagram (same schema, crow's foot notation, with planned MySQL types; main ENCOUNTER columns only):

